Part 3: Gradient Descent Manual Calculation.

We are given.

$$y = mx + b$$

initial values: $m_0 = -1$
$b_0 = 1$
$\alpha = 0.1$

Data points: $(1,3)$ $(3,6)$

number of data $p(n) = 2$

Step 1: Define the Cost function (MSE).

Mean Squared Error is:

$$J(m,b) = \frac{1}{n}\sum (y_i - \hat{y}_i)^2$$

Since $n = 2$

$$J(m,b) = \frac{1}{2}\left[(y_i - \hat{y}_i)^2 + (y_i + \hat{y}_i)^2\right]$$

Step 2: Derive the Gradients.

We compute partial derivatives.

On slope (m)

$$\frac{\partial J}{\partial m} = \frac{-2}{n}\sum x_i (y_i - \hat{y}_i)$$

Since $n = 2$

$$\Rightarrow \frac{\partial J}{\partial m} = -\sum x_i (y_i - \hat{y}_i)$$

On Intercept: b

$$\frac{\partial J}{\partial b} = \frac{-2}{n}\sum (y_i - \hat{y}_i)$$

$n = 2$

$$\Rightarrow \frac{\partial J}{\partial b} = -\sum (y_i - \hat{y}_i)$$

ITERATION 1.

Current Values:

$m = -1,\ b = 1$

Step 1: Compute predictions.

for $x = 1$:

$$\hat{y}_1 = (-1)(1) + 1 = 0$$

for $x = 3$: $\hat{y}_2 = (-1)(3) + 1$
$= -2$

Step 2: Compute Errors.

$$e_1 = 3 - 0 = 3$$
$$e_2 = 6 - (-2) = 8$$

Step 3: Compute Gradients.

for m: $\frac{\partial J}{\partial m} = -[1(3) + 3(8)]$

$= -[3 + 24]$

$$\frac{\partial J}{\partial m} = -24$$

for b:

$$\frac{\partial J}{\partial b} = -[3 + 8]$$

$$\frac{\partial J}{\partial b} = -11$$

Step 4: Update parameters.

$m_1 = -1 - 0.1(-27$

$m_1 = 1.7$

$(m = m_{old} \ldots$

for b

Formula

b_1

b_1

Iter

Curren

Step 1:

$\hat{y}_1 =$

$\hat{y}_2 =$

Step 2

Step

for m:

for b

step.

$m_2 = .$

$m_2 =$

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for b_1

Formula : $b_1 = b_0 + \alpha \frac{2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)$

$b_1 = 1 - 0.1 (-11)$

$b_1 = 2 \cdot 1$

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2ND ITERATION

MSE:

$J(m,b) = (1/n) \times \Sigma (y_1 - y')^2$

where $y = mx + b$

$\therefore$ Let $u = mx + b - y \rightarrow J = (1/n) \times \Sigma u^2$

Partial deriv w respect to m (Chain rule):

$$\frac{dj}{dm} = \frac{dj}{du} \times \frac{du}{dm}$$

$$\frac{dj}{du} = (1/n) \times 2u = 2u/n$$

$$\frac{du}{dm} = x$$

$$\therefore \frac{dj}{jm} = (2u/n) \times x = 2(mx+b-y) \times x/n$$

Expanding for n=2:

$$\frac{dj}{dm} = [(mx_1+b-y_1) \times x_1 + (mx_2+b-y_2) \times x_2]$$

Partial derivative w resp to b:

$$\frac{dj}{db} = \frac{dj}{du} \times \frac{du}{db}$$

$$\frac{du}{db} = 1$$

$$\therefore \frac{dj}{db} = (2u/n) \times 1 = 2(mx+b-y)/n$$

Expanding for n=2:

$$[(mx_1+b-y_1) + (mx_2+b-y_2)]$$

Updated rules:

$$\text{new } m = m - \alpha \times \left(\frac{dj}{jm}\right)$$

$$\text{new } b = b - \alpha \times \left(\frac{dj}{db}\right)$$

Current values:

$m = 1.7$, $b = 2.1$

Prediction:

$\hat{y} = 1.7(1) + 2.1 = 3.8$

$\hat{y}_2 = 1.7(3) + 2.1 = 7.2$

ERRORS:

$MSE_1 = 3 - 3.8 = -0.8$

$MSE_2 = 6 - 7.2 = -1.2$

GRADIENTS:

for m $= \frac{dj}{dm} = -[1(-0.8) + 3(-1.2)]$

$= -(-0.8 - 3.6)$

$\frac{dj}{dm} = 4.4$

for b: $\frac{dj}{db} = -(-0.8 - 1.2)$

$\frac{dj}{db} = 2$

Updated values:

$m_2 = 1.2 - 0.7(4.4)$

$\boxed{m_2 = 1.26}$

$b_2 = 2.1 - 0.1(3)$

$= 2.1 - 0.3$

$\boxed{\text{new } b = 1.9}$

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Iteration 3.

Current Value : $m = 1.26,6 = 1.9$

Step 1 : Predictions

$\hat{y}_1 = 1.26 + 1.9 = 3.16$

$\hat{y}_1 = 5.68$

Step 2 : Errors

$l_1 = 3 - 3.16 = 0.16$

$l_2 = 6 - 5.68 = 0.34$

Step 3 : Gradients.

For $m = \frac{dJ}{dm} = -[1(-0.16) + 3(0.32)]$

$= -[0.8]$

$$\boxed{\frac{dJ}{dm} = 0.8}$$

For b : $\frac{dJ}{db} = -[-0.16 + 0.32]$

$= -0.16$

Step 4: Update

$m_3 = 1.26 - 0.1\,(0.8)$

$m_3 = 1.34$

$b_3 = 1.9 - 0.1\,(-0.16)$

$b_3 = 1.916$

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4th Iteration

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- Current values: $m = 1.34$ $b = 1.916$
- Given points: $(1,3)$ & $(3,6)$ $x_1 = 1$ $y_1 = 3$ $x_2 = 3$ $y_2 = 6$ $\alpha = 0.1$

$\hat{y} = mx + b$

→ Predictions: $\hat{y}_1 = (1.34)(1) + 1.916 = 3.256$ $\hat{y}_2 = (1.34)(3) + 1.916 = 5.936$

→ Errors: $e_1 = \hat{y}_1 - y_1 = 3.256 - 3 = 0.256$ $e_2 = \hat{y}_2 - y_2 = 5.936 - 6 = -0.064$

→ Deriving the gradient of the cost func using MSE

$$MSE = \frac{1}{n}\sum(\hat{y}-y)^2 = \frac{1}{2}(0.256)^2 + (-0.064)^2 = \frac{0.065536 + 0.004096}{2} = \frac{0.069632}{2} = 0.034816$$

$$\hat{y} = mx + b \xrightarrow{\text{substitute}} MSE = \frac{1}{n}\sum(mx+b-y)^2$$

Let $u = mx + b - y$

$$MSE = \frac{1}{n}\sum(mx+b-y)^2 \rightarrow MSE = \frac{1}{n}\sum u^2$$

Differentiating with respect to m

$$\frac{\partial MSE}{\partial m} = \frac{1}{n}\sum \frac{d}{dm}(u^2)$$

$$\frac{\partial MSE}{\partial m} = \frac{\partial MSE}{\partial u} \cdot \frac{\partial u}{\partial m} \rightarrow \text{Chain rule}$$

$$\frac{\partial MSE}{\partial u} = \frac{1}{n}\sum 2u \quad \& \quad \frac{\partial u}{\partial m} = x$$

$$\therefore \frac{\partial MSE}{\partial m} = \frac{1}{n}\sum 2ux$$

With respect to b

$$\frac{\partial MSE}{\partial b} = \frac{1}{n}\sum \frac{d}{db}(u^2)$$

$$\frac{\partial MSE}{\partial b} = \frac{\partial MSE}{\partial u} \cdot \frac{\partial u}{\partial b}$$

$$\frac{\partial MSE}{\partial u} = \frac{1}{n}\sum 2u \quad \& \quad \frac{\partial u}{\partial b} = 1$$

$$\therefore \frac{\partial MSE}{\partial b} = \frac{1}{n}\sum 2u$$

Substituting back 'u' → $\frac{\partial MSE}{\partial m} = \frac{2}{n}\sum(mx+b-y)x$

→ $\frac{\partial MSE}{\partial b} = \frac{2}{n}\sum(mx+b-y)$

$n = 2$

$$\rightarrow \frac{\partial MSE}{\partial m} = \frac{2}{2}(mx_1+b-y_1)x_1 + (mx_2+b-y_2)x_2$$

$= (0.256)1 + (-0.064)3$

$= 0.256 - 0.192 = 0.064$

$$\rightarrow \frac{\partial MSE}{\partial b} = \frac{2}{2}(mx_1+b-y_1) + (mx_2+b-y_2)$$

$= 0.256 - 0.064$

$= 0.192$

→ Updates: $m = m - \alpha\left(\frac{2}{n}\sum(mx_i+b-y_i)x\right)$

$m = 1.34 - 0.1(0.064)$

$= 1.34 - 0.0064 = 1.3336$

$b = b - \alpha \cdot \frac{2}{n}\sum(mx_i - b - y_i)$

$= 1.916 - 0.1(0.192)$

$= 1.916 - 0.0192 = 1.8968$

Final values: $MSE = 0.034816$, $\frac{\partial J}{\partial m} = 0.064$, $\frac{\partial J}{\partial b} = 0.192$, $m_{new} = 1.3336$, $b_{new} = 1.8968$

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→ Trend observation

- The MSE drops significantly over the iterations, confirming that gradient descent is minimizing the error.
- Both m & b fluctuate slightly before settling towards their optimal values.
- The gradients decrease in magnitude with each iteration, indicating the algorithm is approaching the minimum of the cost function & predictions are becoming more accurate.

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